Computing Science (CMPUT) 455

Search, Knowledge, and Simulations

Martin Müller

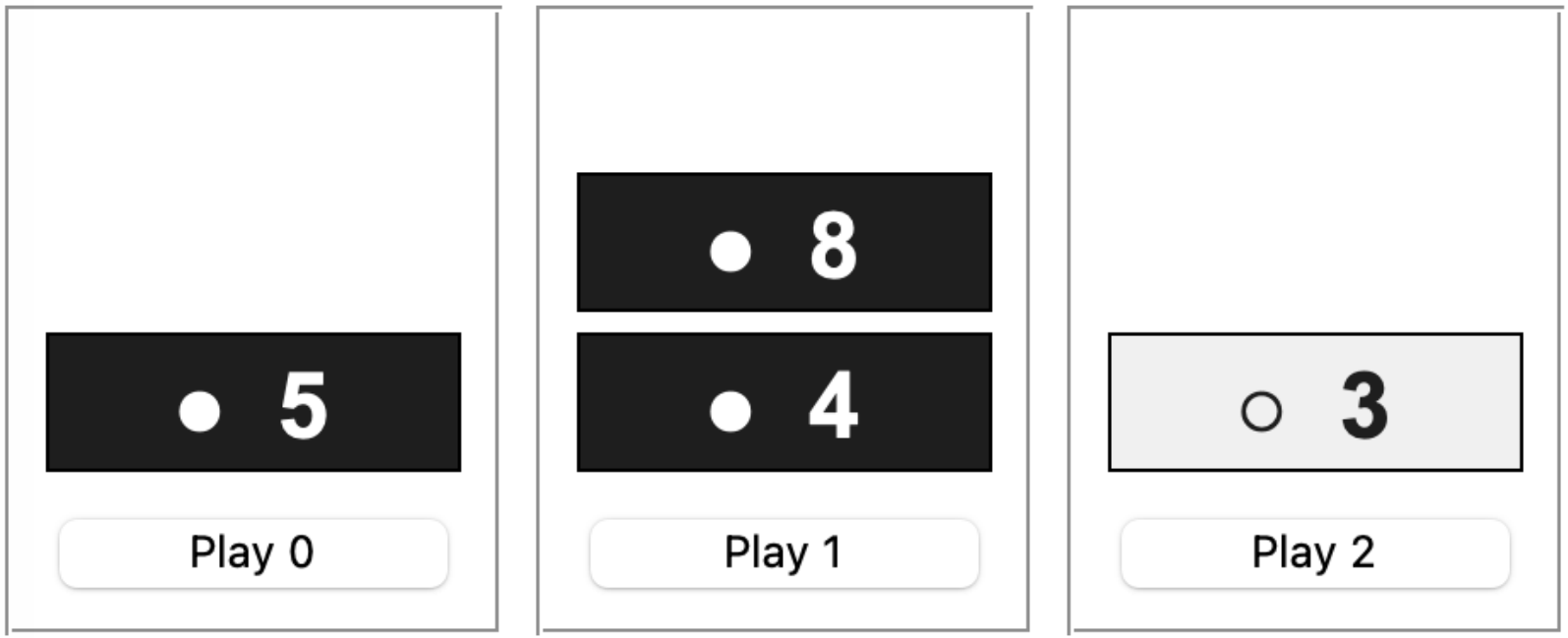
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Assignment 3 Preview

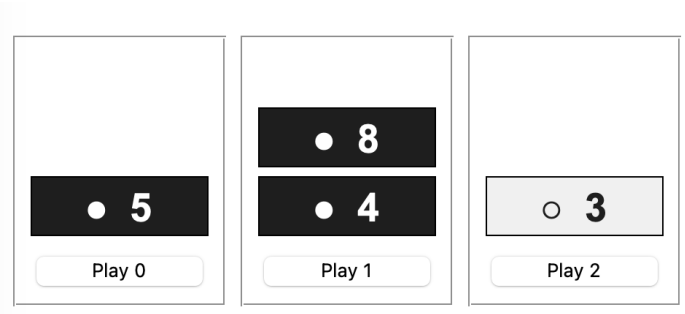
- Goals:
  - Write a **simulation-based player** for Heap Go
  - Implement pure random playouts
  - Implement probabilistic random playouts
  - Use it to `play` Heap Go
- Assignment specification: Canvas
- We will talk about the concepts and algorithms in class starting today

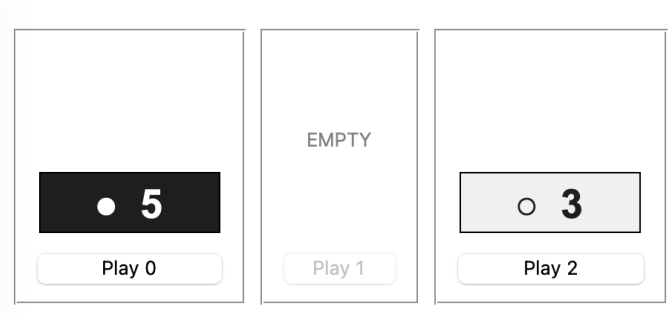
Example: Pure Random Policy

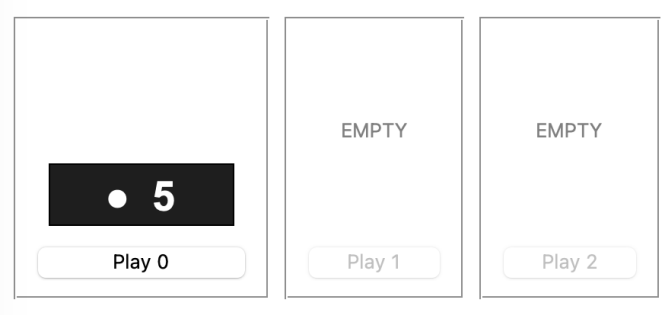


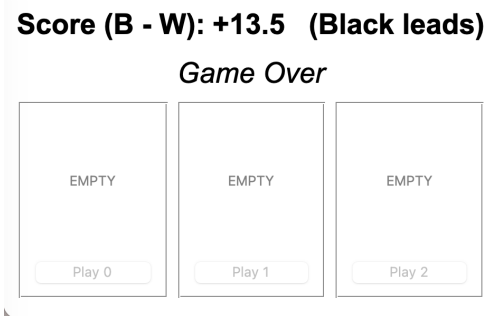
- Black to play, komi 0.5
- 3 legal moves: heaps 0, 1, 2
- Each move chosen with probability 1/3
- Simulate whole game: redo selection after each move

Pure Random Policy - Rest of Game



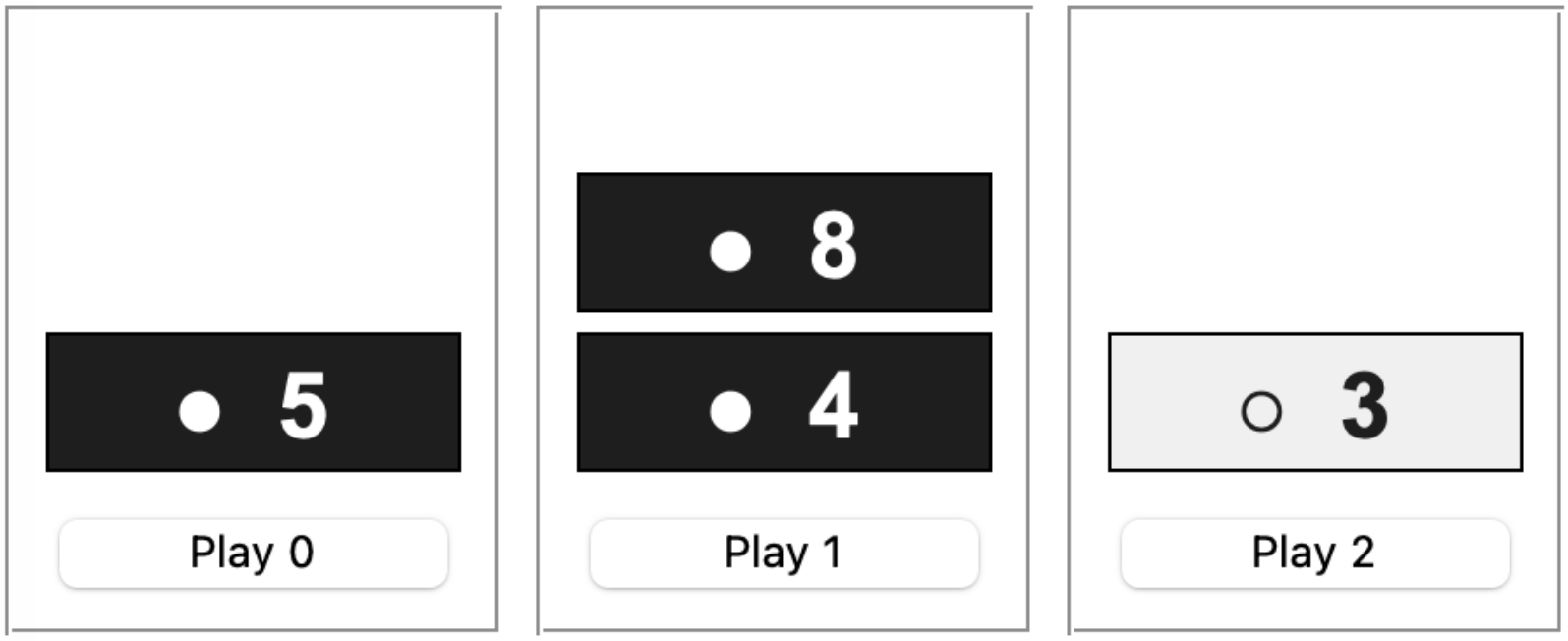






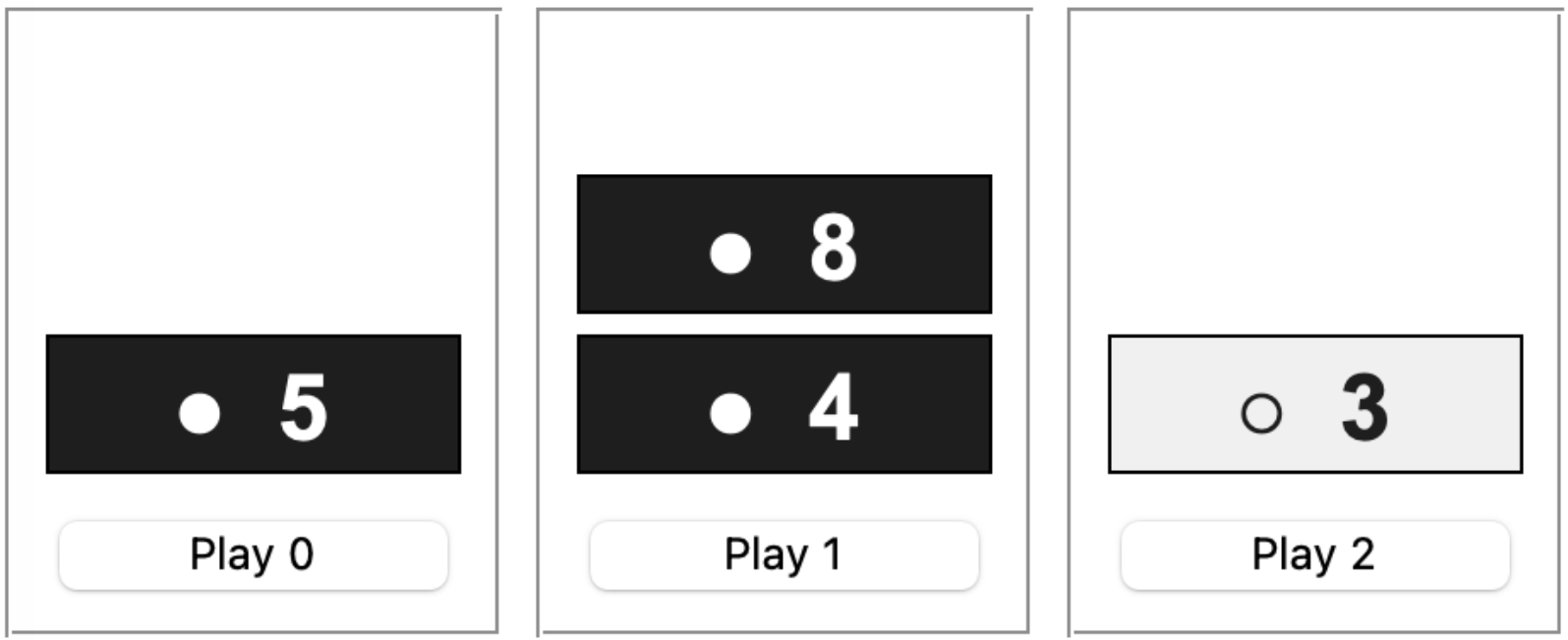
Move 1: legal 0,1,2, prob = 1/3 each, randomly chose heap 1
Move 2: legal 0,2, prob = 1/2 each, randomly chose heap 2
Move 3: legal 0 prob = 1, “randomly” chose heap 0
Simulation over, Black won

Example: Probabilistic Policy



- 3 legal moves: heaps 0, 1, 2
- Assume Black to play
- Greedy move values = sum of token values taken with one move
- In example: 5, 8+4 = 12, 3
- Probabilities 5/20, 12/20, 3/20

Example: Probabilistic Policy



- Same example, but White to play
- Greedy values: 5, 8, 3
- Probabilities 5/16, 8/16, 3/16